Question 1

15 points total | **Distribution of points**

(a) 4 points

For any statement indicating that impulse equals change in momentum	1 point
For any statement indicating that impulse can be determined from the area under the graph	1 point

$\Delta p = F\Delta t =$ area under the curve

$$\Delta p = 2\left(\frac{1}{2}\right)\left(0.5\times10^{-3}\ \text{s}\right)\left(10\times10^{3}\ \text{N}\right) = 5\ \text{N}\cdot\text{s}$$

For recognizing that the impulse on the 2 kg cart is negative, and subtracting it from the initial momentum of the cart	1 point

$$p_{2\text{ kg after}} = (2.0\ \text{kg})(3.0\ \text{m/s}) - 5\ \text{N}\cdot\text{s} = 1\ \text{N}\cdot\text{s}$$

$$p_{2\text{ kg after}} = (2\ \text{kg})\,\upsilon_{2\text{ kg after}} = 1\ \text{N}\cdot\text{s}$$

For the correct answer	1 point

$\upsilon_{2\text{ kg after}} = 0.5\ \text{m/s}$ to the right

(b) 3 points

For any statement of conservation of momentum	1 point

$$p_{2\text{ kg before}} = p_{2\text{ kg after}} + p_{m\text{ after}}$$

For correct substitutions	1 point

$$6\ \text{N}\cdot\text{s} = 1\ \text{N}\cdot\text{s} + m(1.6\ \text{m/s})$$

For the correct answer	1 point

$m = 3.1$ kg

Alternate solution | *Alternate points*

The average acceleration is the average force divided by the mass

$$\bar{a} = \bar{F}/m$$

For calculating the average acceleration	*1 point*

$$\bar{a} = \frac{\Delta\upsilon}{\Delta t} = \frac{1.6\ \text{m/s}}{1\times10^{-3}\ \text{s}} = 1600\ \text{m/s}^2$$

For calculating the average force	*1 point*

$$\bar{F} = \frac{1}{2}(0 + 10{,}000)\ \text{N} = 5000\ \text{N}$$

$$m = (5000\ \text{N})/(1600\ \text{m/s}^2)$$

For the correct answer	*1 point*

$m = 3.1$ kg

Note: An alternate solution is to do part (b) first using impulse, in which case the first two points noted above for part (a) could be earned. Then conservation of momentum can be used to solve part (a), so the first point noted above for part (b) could be earned.

Question 1 (cont'd.)

	Distribution of points
(c) 2 points	
For using the slope of the graph to determine the acceleration	1 point
$a = \text{ slope } = \dfrac{(0.5 - 1.6)\ \text{m/s}}{(3.5 - 2)\ \text{s}}$	
For the correct answer	1 point
$a = -0.73\ \text{m/s}^2$	
(d) 3 points	
For using the area under the curve or one of the equations $d = vt$ or $d = \bar{v}t$	1 point
For calculating a distance for each segment of the graph	1 point
$d_1 = (1.6\,\text{m/s})(2\ \text{s}) = 3.2\ \text{m}$	
$d_2 = \dfrac{(1.6 + 0.5)\ \text{m/s}}{2}(1.5\ \text{s})$ <u>or</u> $(1.6\ \text{m/s})(1.5\ \text{s}) + \dfrac{1}{2}\left(-0.73\ \text{m/s}^2\right)(1.5\ \text{s})^2$	
$d_2 = 1.6\ \text{m}$	
$d_3 = (0.5\ \text{m/s})(1.5\ \text{s}) = 0.8\ \text{m}$	
$d_{\text{tot}} = d_1 + d_2 + d_s$	
For the correct answer	1 point
$d_{\text{tot}} = 5.5\ \text{m}$	
(e) 3 points	
The acceleration is negative, so the cart must be moving opposite to the force of gravity, which is the only force acting on it.	
For indicating that the ramp goes up	1 point
For using conservation of energy	1 point
$mgh = \dfrac{1}{2}m\left(v_i^2 - v_f^2\right)$	
$h = \dfrac{1}{2\left(10\ \text{m/s}^2\right)}\left((1.6\ \text{m/s})^2 - (0.5\ \text{m/s})^2\right)$	
For the correct answer	1 point
$h = 0.12\ \text{m}$	